

Linear Algebra - section 1.5

Solution sets of Linear Systems

- Def: A system of linear equations is homogeneous if it can be written in the form $A\vec{x} = \vec{0}$.

* Homogeneous systems always have at least one solution: $\vec{x} = \vec{0}$ (called the trivial solution).

* We want to know if there are non zero solutions.

- Ex:
$$\begin{cases} 2x_1 - 5x_2 + 8x_3 = 0 \\ -2x_1 - 7x_2 + x_3 = 0 \\ 4x_1 + 2x_2 + 7x_3 = 0 \end{cases}$$
 ← does this have a nonzero solution?

$$\left[\begin{array}{ccc|c} 2 & -5 & 8 & 0 \\ -2 & -7 & 1 & 0 \\ 4 & 2 & 7 & 0 \end{array} \right]$$

$$\frac{1}{2}R_1 \rightarrow R_1 \left[\begin{array}{ccc|c} 1 & -5/2 & 4 & 0 \\ -2 & -7 & 1 & 0 \\ 4 & 2 & 7 & 0 \end{array} \right]$$

$$\begin{array}{l} 2R_1 + R_2 \rightarrow R_2 \\ -4R_1 + R_3 \rightarrow R_3 \end{array} \left[\begin{array}{ccc|c} 1 & -5/2 & 4 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 12 & -9 & 0 \end{array} \right]$$

$$R_2 + R_3 \rightarrow R_3 \left[\begin{array}{ccc|c} 1 & -5/2 & 4 & 0 \\ 0 & -12 & 9 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$-1/12 R_2 \rightarrow R_2 \quad \left[\begin{array}{ccc|c} 1 & -5/2 & 4 & 6 \\ 0 & 1 & -3/4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$$5/2 R_2 + R_1 \rightarrow R_1 \quad \left[\begin{array}{ccc|c} 1 & 0 & 17/8 & 0 \\ 0 & 1 & -3/4 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

x_3 is free (row of zeros)

$$x_2 - 3/4 x_3 = 0 \rightarrow x_2 = 3/4 x_3$$

$$x_1 + 17/8 x_3 = 0 \rightarrow x_1 = -17/8 x_3$$

* we can write our answer in parametric form:

$$\vec{x} = \begin{bmatrix} -17/8 t \\ 3/4 t \\ t \end{bmatrix} \quad t \in \mathbb{R}.$$

when $t=8$ we get a nonzero solution:

$$\vec{x} = \begin{bmatrix} -17 \\ 6 \\ 8 \end{bmatrix}$$